AI MEDSCOPE: IN DEPTH STUDY ON THE REAL WORLD APPLICATIONS OF ARTIFICIAL INTELLIGENCE IN MODERN HEALTHCARE

A "Project Stage II" Report submitted to
JNTU Hyderabad in partial fulfillment
of the requirements for the award of the degree

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND ENGINEERING

Submitted by

JUPAKA RAJESH	**22S11A0596**
BOINI SHIVA KUMAR	**22S11A05B2**
MUKTHALA SUKENDER	**22S11A05B9**
MERIGA THARUN	**22S11A05C3**

Under the Guidance of

Mr. BHASKER REDDY
B.Tech, M.Tech,
Professor of CSE

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
MALLA REDDY INSTITUTE OF TECHNOLOGY & SCIENCE
(Approved by AICTE New Delhi and Affiliated to JNTUH)
(Accredited by NBA & NAAC with "A" Grade)
An ISO 9001: 2015 Certified Institution
Maisammaguda, Medchal (M), Hyderabad-500100, T. S.

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CERTIFICATE

This is to certify that the "Project Stage II" entitled "AI MEDSCOPE : IN DEPTH STUDY ON THE REAL WORLD APPLICATIONS OF ARTIFICIAL INTELLIGENCE IN MODERN HEALTH CARE" has been submitted by JUPAKA RAJESH (22S11A0596), BOINI SHIVA KUMAR(22S11A05B2),MUKTHALASUKENDER(22S11A05B9)and MERIGA THARUN (22S11A05C3) in partial fulfillment of the requirements for the award of **BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE & ENGINEERING.** This record of bonafide work carried out by them under my guidance and supervision. The result embodied in this Project Stage II report has not been submitted to any other University or Institute for the award of any degree.

Mr. Bhasker Reddy
Professor of CSE &
Project Guide

Mrs. K Mamatha
Head of the Department
of CSE

External Examiner

ACKNOWLEDGEMENT

The Project Stage II work carried out by our team in the Department of Computer Science and Engineering, Malla Reddy Institute of Technology and Science, Hyderabad. ***This work is original and has not been submitted in part or full for any degree or diploma of any other university.***

We wish to acknowledge our sincere thanks to our project guide **Dr.Vaka Murali Mohan**, Principal and Professor of CSE, guidance and her continuous supervision during the course of work.

We acknowledge our sincere thanks to **Dr.Vaka Murali Mohan,** Principal and **Mrs. K Mamatha** Head of the Department and Coordinator, faculty members of CSE Department for their kind cooperation in making this Project Stage II work a success.

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JUPAKA RAJESH	**22S11A0596**	_____
BOINI SHIVA KUMAR	**22S11A05B2**	_____
MUKTHALA SUKENDER	**22S11A05B9**	_____
MERIGA THARUN	**22S11A05C3**	_____

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ABSTRACT

With the advancement of science and technology, the application of artificial intelligence is of great significance in various fields, and it has been a huge driving force in the development of more and more fields. Through research on the application of artificial intelligence in distance medical teaching, virtual inquiry, distance education management, teaching video recording, etc., this article concludes that artificial intelligence can increase the efficiency of medical teaching, improve visual utility, and think more like human, thus it can better serve the people. The application effects of artificial intelligence in the field of medical education, especially for the improvement of the overall quality of medical students, provide much inspiration for the applications of artificial intelligence in medical education.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
PACS	Picture Archieving and Communication System
UML	Unified Modeling Diagram
DT	Decision Tree Algorithm
CSV	Comma Seperated Values
HTML	Hyper Text Markup Language
HTTP	Hyper Text Transfer Protocol
DFD	Data Flow Diagram

CHAPTER-1: SYSTEM ANALYSIS

1.1 Existing System

Computer technology could be used to reduce the number of mortality and reduce the waiting time to see the specialist. Computer program or software developed by emulating human intelligence could be used to assist the doctors in making decision without consulting the specialists directly. The software was not meant to replace the specialist or doctor, yet it was developed to assist general practitioner and specialist take an immediate action to produce as many doctors as possible. However, while waiting for students to become doctors and the doctors to become specialists, many patients may already die. Current practice for medical treatment required patients to consult specialist for further treatment. Artificial intelligence provides students with more opportunities to participate in a digital and dynamic way.

1.1.1 Disadvantages of Existing System

1. High Waiting Time: Patients still need to wait a long time to consult specialists, leading to delays in diagnosis and treatment.

2. Limited Access to Specialists: In many areas, there is a shortage of specialists, which makes it difficult for patients to receive timely expert care.

3. Increased Mortality Rate: Due to the delay in specialist consultation, patients with critical conditions may not receive immediate attention, increasing the risk of death.

4. Dependence on Human Expertise: The system heavily relies on doctors and specialists, making it less efficient when there is a shortage of trained professionals.

5. Slow Decision-Making Process: Without AI support, doctors must manually analyze patient data, which takes more time and may lead to delayed decisions.

1.2 Proposed System

- Through modern information technology, data centers, teaching resources library, cloud platform, are constructed for students recruiting, training process management and evaluation, which can improve the efficiency and service level of continuing medical education management. In the sharing management of the base, institutional management, trainees, project management, evaluation, credit management and teaching content such as artificial intelligence in distance medical teaching, virtual inquiry, distance education management, teaching video recording, etc., especially for the improvement of the overall quality of medical students, provide much inspiration for the applications of artificial intelligence in medical education.

1.2.1 Advantages of Proposed System

1. Improved Learning Efficiency: The integration of cloud platforms and AI tools makes medical education faster, more organized, and efficient for both students and educators.

2. Easy Access to Learning Resources: Students can access digital libraries, video lectures, and virtual simulations anytime and anywhere, promoting flexible and continuous learning.

3. Enhanced Quality of Medical Training: Artificial intelligence provides personalized learning experiences and virtual medical training, helping students understand complex medical concepts better.

4. Effective Management and Evaluation: AI-based systems simplify student recruitment, training process management, performance tracking, and evaluation, improving overall administrative efficiency.

5. Real-Time Data Sharing: Cloud technology enables seamless sharing of educational materials, records, and updates among institutions, teachers, and trainees.

1.2 Introduction

As humans place great hopes on artificial intelligence, artificial intelligence can have bright prospects, and it may be applied to all aspects of our lives to improve the overall standard of living of all of us. With the development of artificial intelligence technology, scientists have gradually applied artificial intelligence technology to the teaching field, and quite good results have been achieved. Therefore, the development and progress in artificial intelligence, combined with teaching, will be an excellent new teaching method.The first one is the ability of "natural language processing" and to write computer programs that can be read and spoken. The second one is to develop a machine that has sensitive sensory, and can simulate human hearing and vision and distinguish different environments automatically. The third type is an R&D expert system that uses a computer to simulate an expert's behavior.

Artificial intelligence provides students with more opportunities to participate in a digital and dynamic way. These opportunities are often not found in outdated textbooks or the fixed environment of the classroom. Through this collaborative learning method, each student has the potential to advance others, and can accelerate the exploration of new learning and the creation of innovative technologies. Four applications are provided below to illustrate how artificial intelligence can be applied to medical education. DxR Clinician is an online virtual patient system that uses artificial intelligence technology specifically for teaching hospitals, medical colleges, and residents. The system is widely used in education and clinical thinking evaluation of medical students. The software collects hundreds of real patient data and is compiled by experts and artificial intelligence as specific cases. These cases cover a wide range of clinical issues. Medical students make diagnoses through inquiry, simulated physical examinations, and supplementary examinations of virtual patients to diagnose and provide treatment plans. For teachers, DxR Clinician can be used as a useful analysis tool to help teachers understand students' behavior and adjust courses through appraisal results. For students, they can quickly develop clinical problem solving skills. By interacting with the cases, students can learn a lot about important disease diagnosis. At the same time, the system can identify mistakes that students make in the process of case analysis, conducts deep learning and analysis, and help students solve these problems. One kind of computer software that has similar function with DxR Clinician is called Intelligent Tutor Systems, which can track the learner's "psychological steps" in the process of solving problems to diagnose the wrong concepts and estimate the learners' understanding extent of the field. The Intelligent Tutor System can also provide learners with timely guidance, feedback and explanation, and can promote learners' learning behaviors such as self-regulation, self-monitoring and self-explaining. Distance education is a kind of teaching method that is not limited by time and space and can

realize real-time on-line and off-line teaching. Learning, communication and sharing can be conducted through web-based teaching methods such as microblogging; virtual simulation training, mobile ward round in clinical practice teaching and mobile nursing play n important role in medical teaching, especially virtual simulation teaching technology has gained more in-depth and extensive application; the development of remote transmission technology of imaging and pathological films, instant transfer technology, all online storage technology, active monitoring and self-healing technology, integrated platform technology, three-dimensional post-processing, computer-aided diagnosis, and medical imaging real-time conferencing technology have had a profound impact on the teaching methods; regional Picture Archiving and Communication System (PACS) and regional pathology platform

CHAPTER 2 LITERATURE SURVEY

1. Learning Environments and Displacement of Learning [J]. British Journal of Educational Technology,2010

AUTHORS : Thomas H. Learning Spaces,

Traditionally, at least according to popular wisdom, learning took place in venues that were custom-designed for the purpose. The purpose, given the evidence of the artefacts with which we are confronted, seems to have been the educational equivalent of the production line that so succinctly characterised the industrialisation of society

2. Use of Information Technology in Medical Education [J]. Webmed Central Medical Education,2010

AUTHORS: Department of Respiratory Medicine and Allergy, Guy's Hospital, London,

The past few years have seen rapid advances in communication and information technology (C&IT), and the pervasion of the worldwide web into everyday life has important implications for education.

3. Improving the Quality of University Teaching in Learning Innovation: The Key to Apply Information Technology in Higher Education,2012

AUTHORS: Xinmin Sang, Yangbin Xie

Information and communication technologies (ICT) have become commonplace entities in all aspects of life. Across the past twenty years the use of ICT has fundamentally changed the practices and procedures of nearly all forms of endeavour within business and governance.

4. Educational Ecosystem in the View of New Industrial Revolution,2015

AUTHORS : Youmei Wang. Cultivating Makers

Higher education in the fourth industrial revolution (HE 4.0) is a complex, dialectical and exciting opportunity which can potentially transform society for the better. The fourth industrial revolution is powered by artificial intelligence and it will transform the workplace from tasks based characteristics to the human centred characteristics

5. NMC Horizon Report 2016 Higher Education Edition,2016

AUTHORS: Johnson L, Adams Becker S, Cummins M.

Nowadays Smart phones have become an irresistible part of everyone’s life. With these smart phones the human life is changed for better. “Android Based Campus Solution” is a college management application which basically aimed at managing most of the activities of college.

CHAPTER-3 SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE:



Fig-1 System architecture

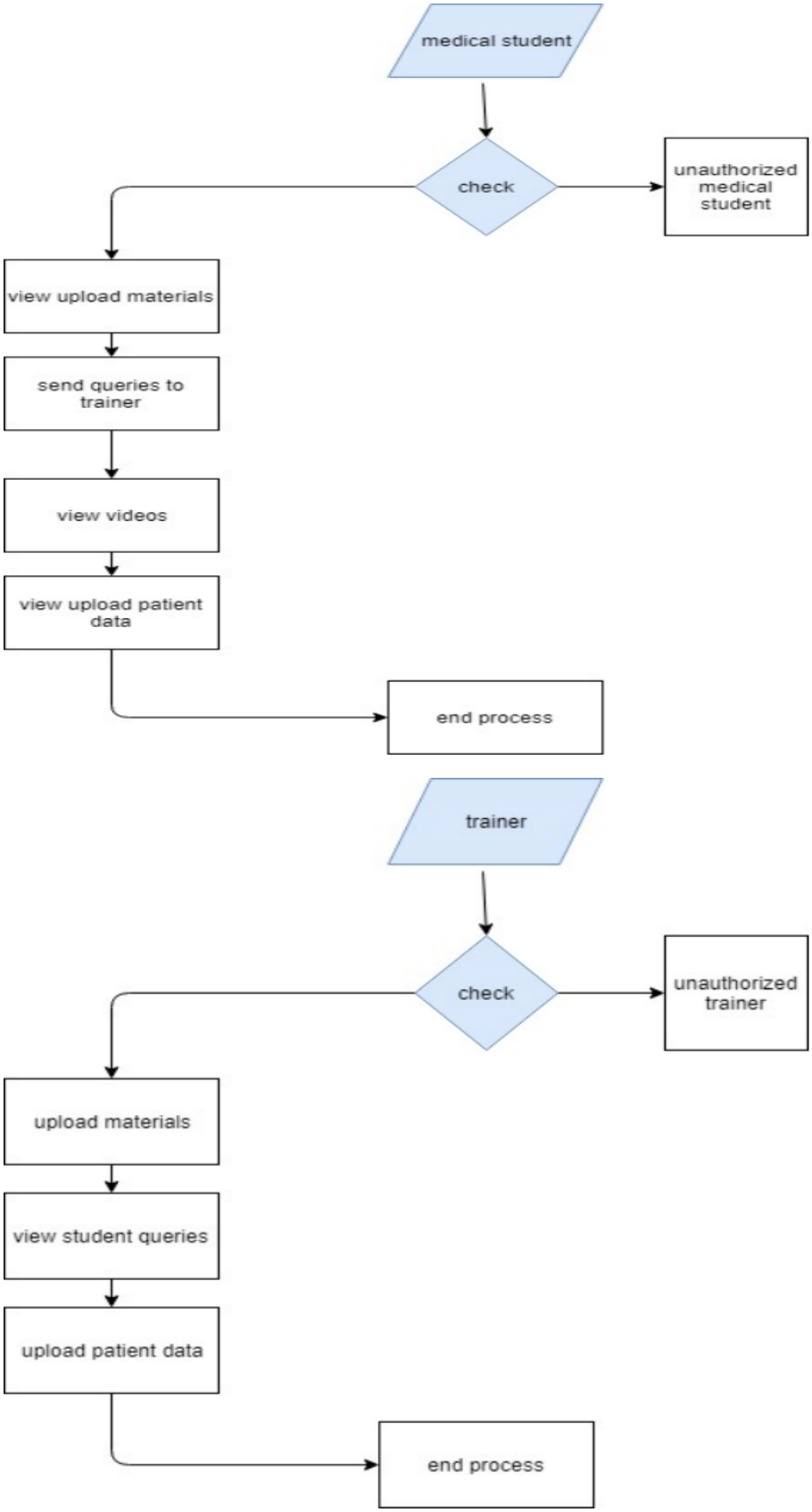
3.2 DATA FLOW DIAGRAM:

1. The DFD is also called as bubble chart. It is a simple graphical formalism that can be used to represent a system in terms of input data to the system, various processing carried out on this data, and the output data is generated by this system.

2. The data flow diagram (DFD) is one of the most important modeling tools. It is used to model the system components. These components are the system process, the data used by the process, an external entity that interacts with the system and the information flows in the system.

3. DFD shows how the information moves through the system and how it is modified by a series of transformations. It is a graphical technique that depicts information flow and the transformations that are applied as data moves from input to output.

4. DFD is also known as bubble chart. A DFD may be used to represent a system at any level of abstraction. DFD may be partitioned into levels that represent increasing information flow and functional detail.


medical student
check
unauthorized medical student
view upload materials
send queries to trainer
view videos
view upload patient data
end process
trainer
check
unauthorized trainer
upload materials
view student queries
upload patient data
end process

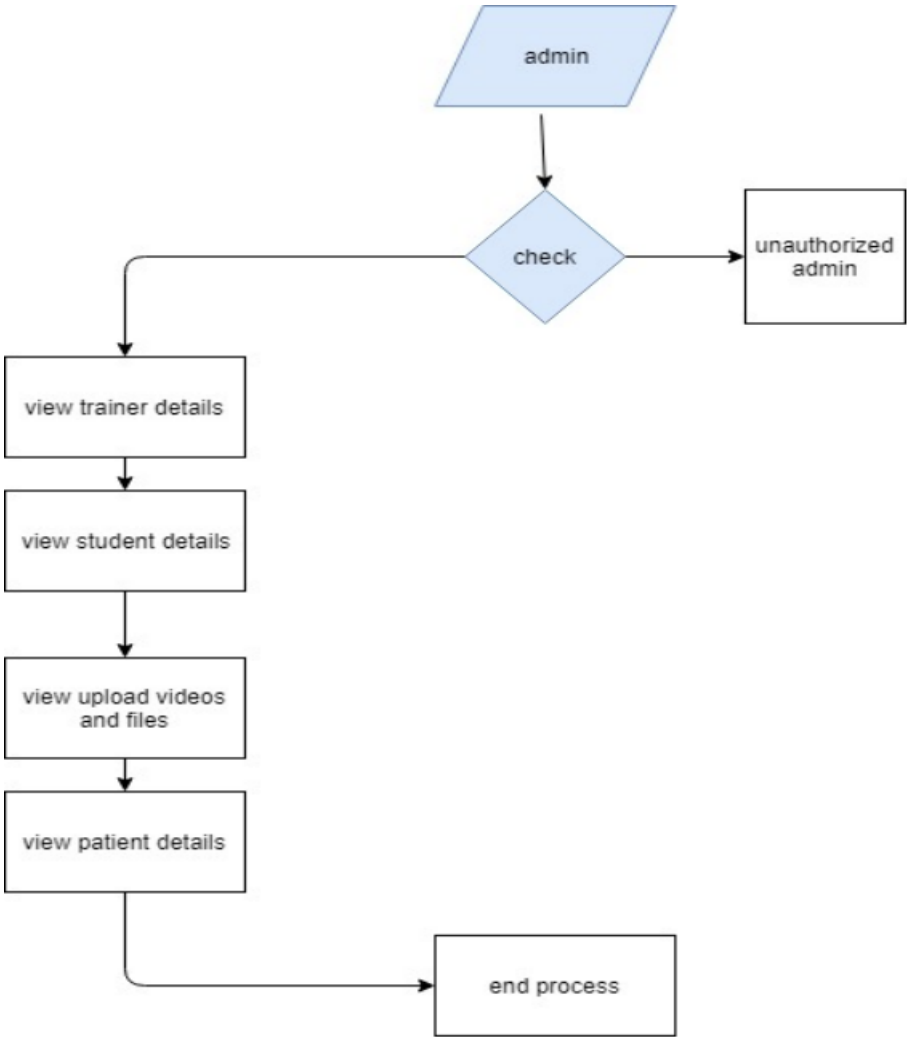


Fig-2 Data flow diagram

3.3 UML DIAGRAMS

UML stands for Unified Modeling Language. UML is a standardized general-purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group.

The goal is for UML to become a common language for creating models of object oriented computer software. In its current form UML is comprised of two major components: a Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML.The Unified Modeling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modeling and other non-software systems. The UML

represents a collection of best engineering practices that have proven successful in the modeling of large and complex systems.The UML is a very important part of developing objects oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

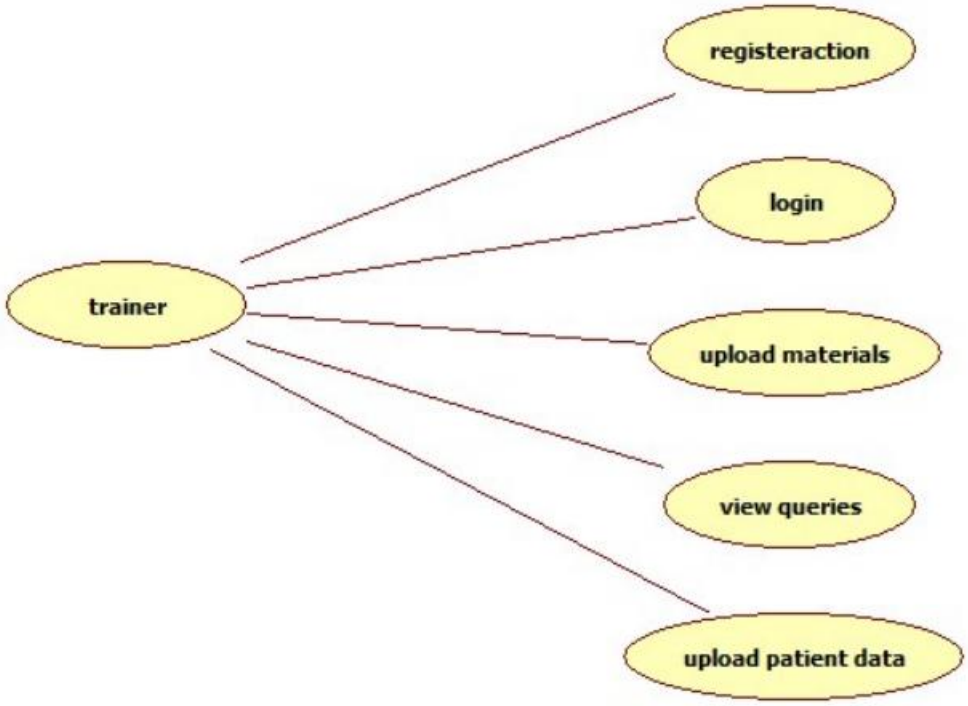
3.4 GOALS:

The Primary goals in the design of the UML are as follows:

1. Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.
2. Provide extendibility and specialization mechanisms to extend the core concepts.
3. Be independent of particular programming languages and development process.
4. Provide a formal basis for understanding the modeling language.
5. Encourage the growth of OO tools market.
6. Support higher level development concepts such as collaborations, frameworks, patterns and components.
7. Integrate best practices.

3.5 USE CASE DIAGRAM:

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.



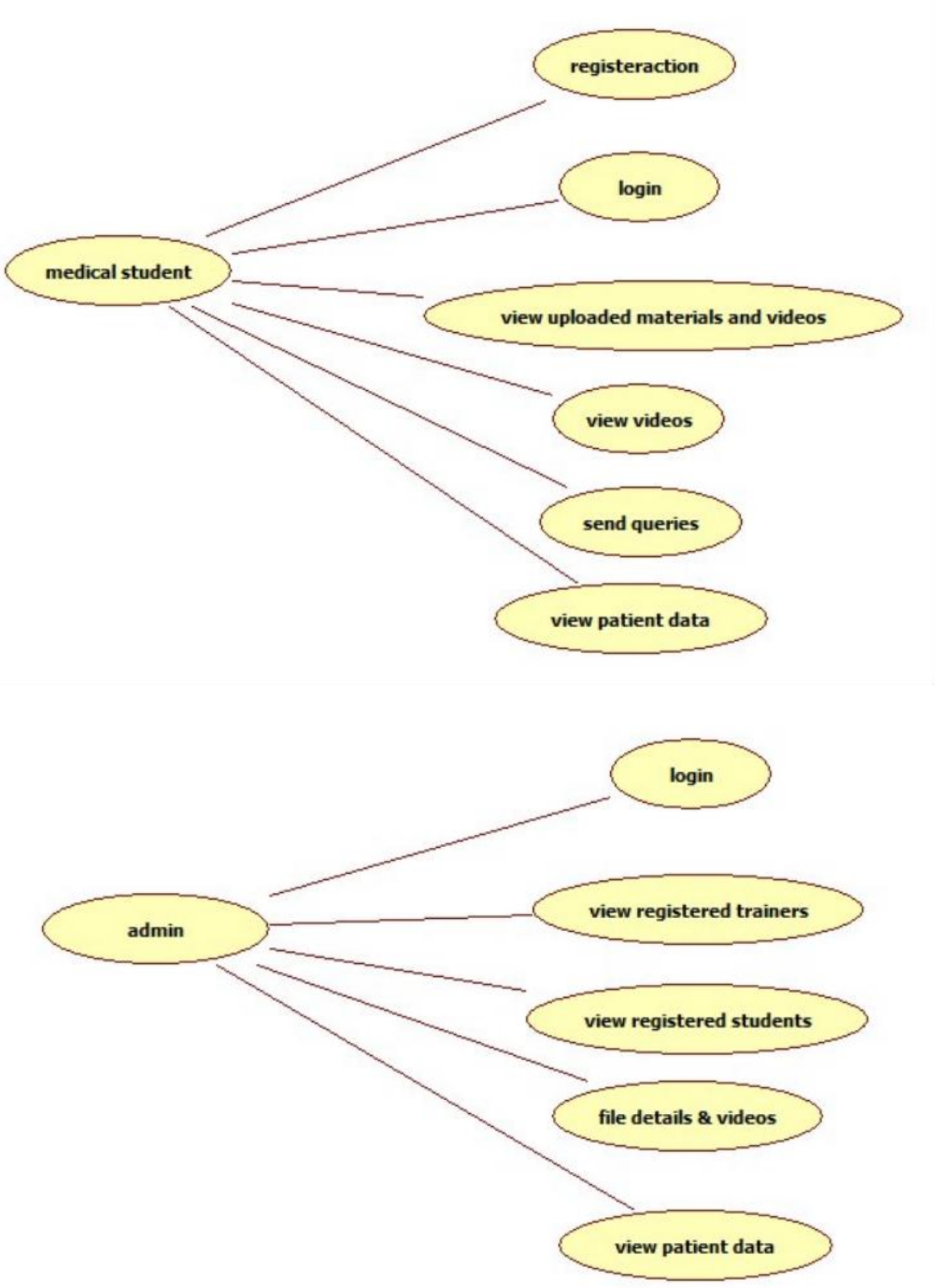


Fig-3 Use case diagram

3.6 CLASS DIAGRAM:

In software engineering, a class diagram in the Unified Modeling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among the classes. It explains which class contains informati

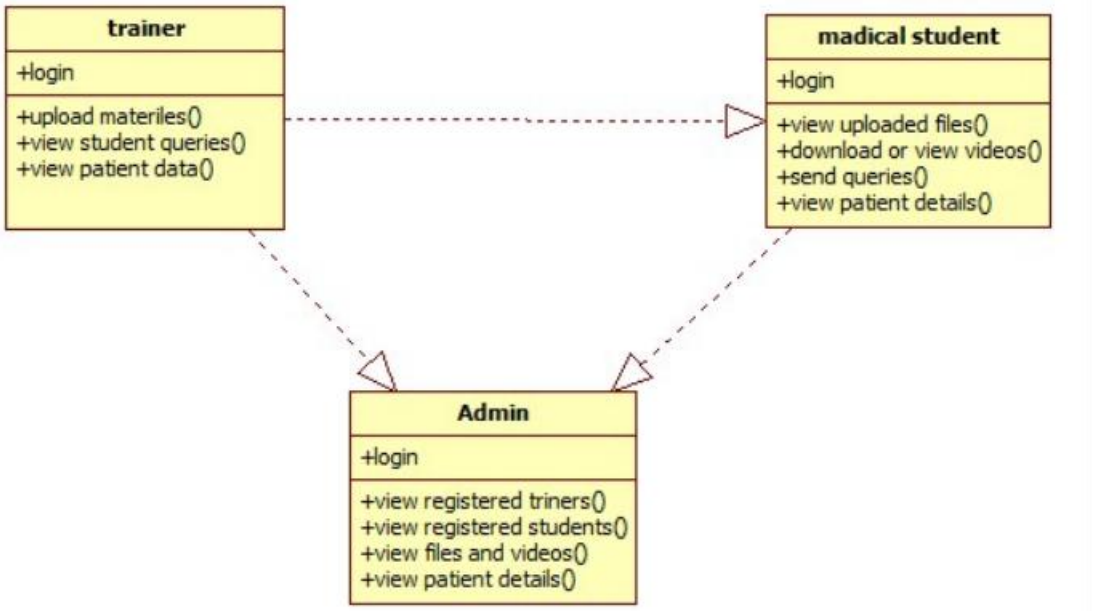


Fig-4 Class diagram

3.7 SEQUENCE DIAGRAM:

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams.

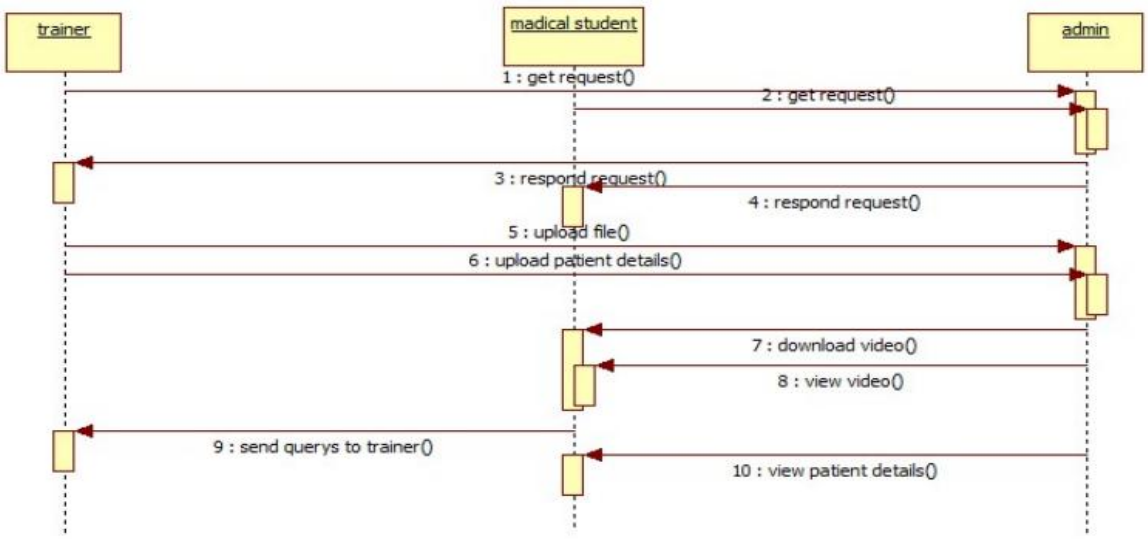


Fig-5 Sequence Diagram

3.8 ACTIVITY DIAGRAM:

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the Unified Modeling Language, activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system. An activity diagram shows the overall flow of control.

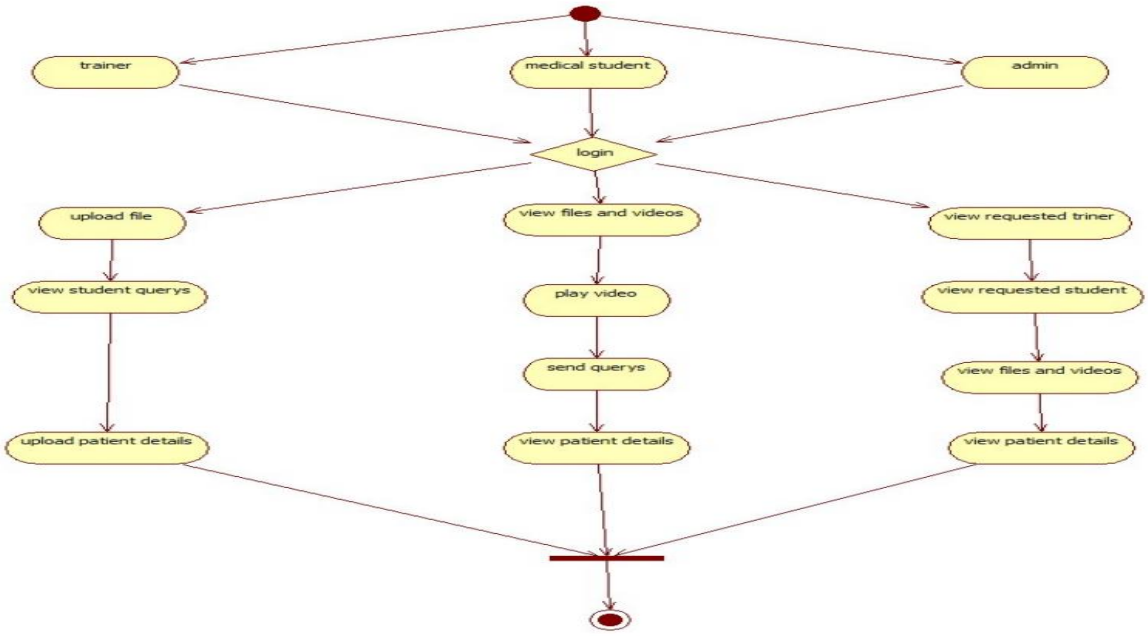


Fig-6 Activity Diagram

3.9 MODULES:

- TRAINER
- MEDICAL STUDENT
- ADMIN
- ARTIFICIAL INTELLIGENCE

TRAINER

In residency training, in the process of students' recruitment, announcement, acceptance, teachers' teaching and courses set-up, the common parts can be exchanged and coordinated through the computer system. Among base hospitals, health administrative departments, clinical teachers and departments, the same information can be transmitted through information means help students at different levels, and provide help and support when teachers and students need it. Artificial intelligence can not only help teachers and students design courses that meet their needs, but also can focus on student performance and alert teachers when problems may arise, helping teachers improve teaching methods. Artificial intelligence will change the role of teachers. Teachers will supplement artificial intelligence courses to provide students with interpersonal interaction and practical experience.

MEDICAL STUDENT

Artificial intelligence provides students with more opportunities to participate in a digital and dynamic way. These opportunities are often not found in outdated textbooks or the fixed environment of the classroom. Through this collaborative learning method, each student has the potential to advance others, and can accelerate the exploration of new learning and the creation of innovative technologies. Four applications are provided below to illustrate how artificial intelligence can be applied to medical education. Virtual Inquiry System, Medical Distance Learning, Influence of Artificial Intelligence Technology on the Management of Distance Medical Education, Recording teachers' teaching videos.

ADMIN:

Admin gives the activation permission to users , admin can verify the trainer and student status. the admin can calculate raw SVM support vector machine. the admin can also calculate the raw svm DT (decision tree algorithm). Window-based anomaly detection divides time-series instances into overlapping windows. Anomaly scores are first calculated per window, and then aggregated for comparison with a predefined

threshold. In supervised prediction-based anomaly detection, a machine-learning-based predictive model is first learned from historical logs The accumulated difference between these predicted and the actual observations is defined as the anomaly score for each test time-series instance.

ARTIFICIAL INTELLIGENCE:

Artificial intelligence is a new science that emerged in the middle of the 20th century. This science mainly belongs to computer science, but it covers information science, linguistics, psychology, philosophy, mathematics, and many other disciplines. It is a discipline that has strong comprehensiveness. Artificial intelligence mainly uses computer systems to simulate human thinking activities. This discipline has a wide research scope and it has also been applied in many aspects. Because artificial intelligence has wide research fields, it is also a very challenging science category requiring scientists to have a strong knowledge base in all aspects. At present, the research of artificial intelligence is closely related to the current needs of human beings. The research on artificial intelligence technology has also evolved with the changes of the times, so that the artificial intelligence technology can be applied to more meaningful things. The main goal of artificial intelligence is to require computers to have “abilities to acquire and learn knowledge”, “abilities to process knowledge”, “abilities to understand language”, “the ability to infer and search automatically”, and abilities in many other aspects.

3.10 SYSTEM REQUIREMENTS:

3.10.1 HARDWARE REQUIREMENTS:

- **System** : Pentium IV 2.4 GHz.
- **Hard Disk** : 1 TB.
- **Floppy Drive** : 1.44 Mb.
- **Monitor** : 14’ Colour Monitor.
- **Mouse** : Optical Mouse.
- **Ram** : 4GB.

3.10.2 SOFTWARE REQUIREMENTS:

- **Operating system** : Windows 7 Ultimate.
- **Coding Language** : Python.
- **Front-End** : Python.
- **Designing** : Html,css,javascript.
- **Data Base** : MySQL.

CHAPTER-4 INPUT AND OUTPUT DESIGN

4.1 INPUT DESIGN

The input design is the link between the information system and the user. It comprises the developing specification and procedures for data preparation and those steps are necessary to put transaction data in to a usable form for processing can be achieved by inspecting the computer to read data from a written or printed document or it can occur by having people keying the data directly into the system. The design of input focuses on controlling the amount of input required, controlling the errors, avoiding delay, avoiding extra steps and keeping the process simple. The input is designed in such a way so that it provides security and ease of use with retaining the privacy. Input Design considered the following things:

- What data should be given as input?
- How the data should be arranged or coded?
- The dialog to guide the operating personnel in providing input.
- Methods for preparing input validations and steps to follow when error occur.

OBJECTIVES

- Input Design is the process of converting a user-oriented description of the input into a computer-based system. This design is important to avoid errors in the data input process and show the correct direction to the management for getting correct information from the computerized system
- It is achieved by creating user-friendly screens for the data entry to handle large volume of data. The goal of designing input is to make data entry easier and to be free from errors. The data entry screen is designed in such a way that all the data manipulates can be performed. It also provides record viewing facilities.
- When the data is entered it will check for its validity. Data can be entered with the help of screens. Appropriate messages are provided as when needed so that the user will not be in maize of instant. Thus the objective of input design is to create an input layout that is easy to follow

4.2 OUTPUT DESIGN

A quality output is one, which meets the requirements of the end user and presents the information clearly. In any system results of processing are communicated to the users and to other system through outputs. In output design it is determined how the information is to be displaced for immediate need and also the hard copy output. It is the most important and direct source information to the user. Efficient and intelligent output design improves the system's relationship to help user decision-making.

1. Designing computer output should proceed in an organized, well thought out manner; the right output must be developed while ensuring that each output element is designed so that people will find the system can use easily and effectively. When analysis design computer output, they should Identify the specific output that is needed to meet the requirements.

2.Select methods for presenting information.

3.Create document, report, or other formats that contain information produced by the system.

The output form of an information system should accomplish one or more of the following objectives.

- Convey information about past activities, current status or projections of the
- Future.
- Signal important events, opportunities, problems, or warnings.
- Trigger an action.
- Confirm an action.

CHAPTER-5 SYSTEM ENVIRONMENT

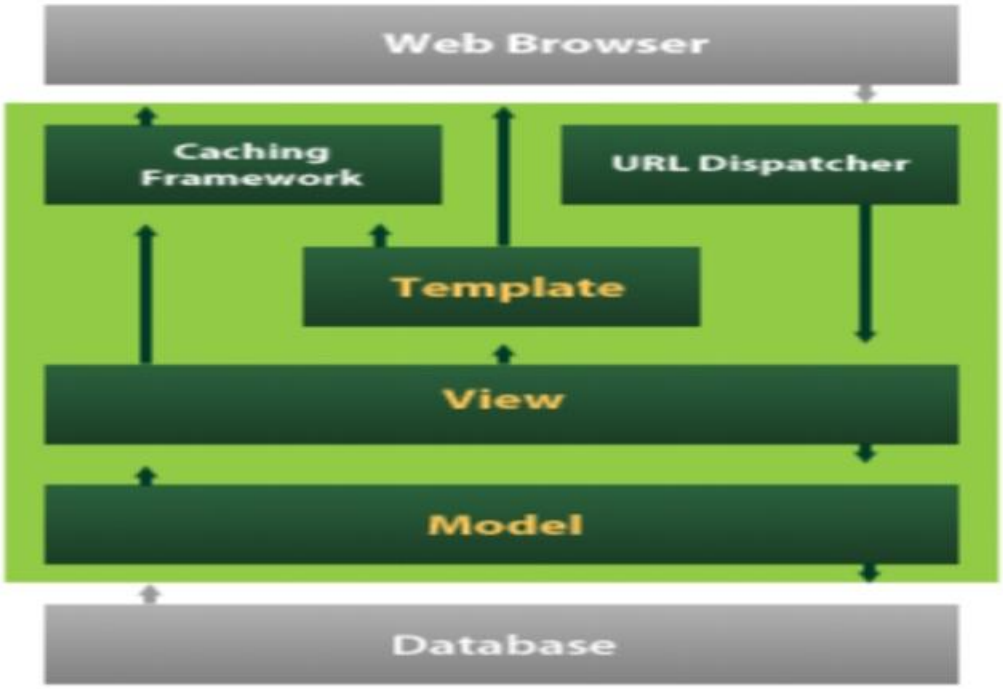
PYTHON

- Python is a general-purpose interpreted, interactive, object-oriented, and high-level programming language. An interpreted language, Python has a design philosophy that emphasizes code readability (notably using whitespace indentation to delimit code blocks rather than curly brackets or keywords), and a syntax that allows programmers to express concepts in fewer lines of code than might be used in languages such as C++or Java.
- It provides constructs that enable clear programming on both small and large scales. Python interpreters are available for many operating systems. CPython, the reference implementation of Python, is open source software and has a community-based development model, as do nearly all of its variant implementations.
- CPython is managed by the non-profit Python Software Foundation. Python features a dynamic type system and automatic memory management. It supports multiple programming paradigms, including object-oriented, imperative, functional and procedural, and has a large and comprehensive standard library.

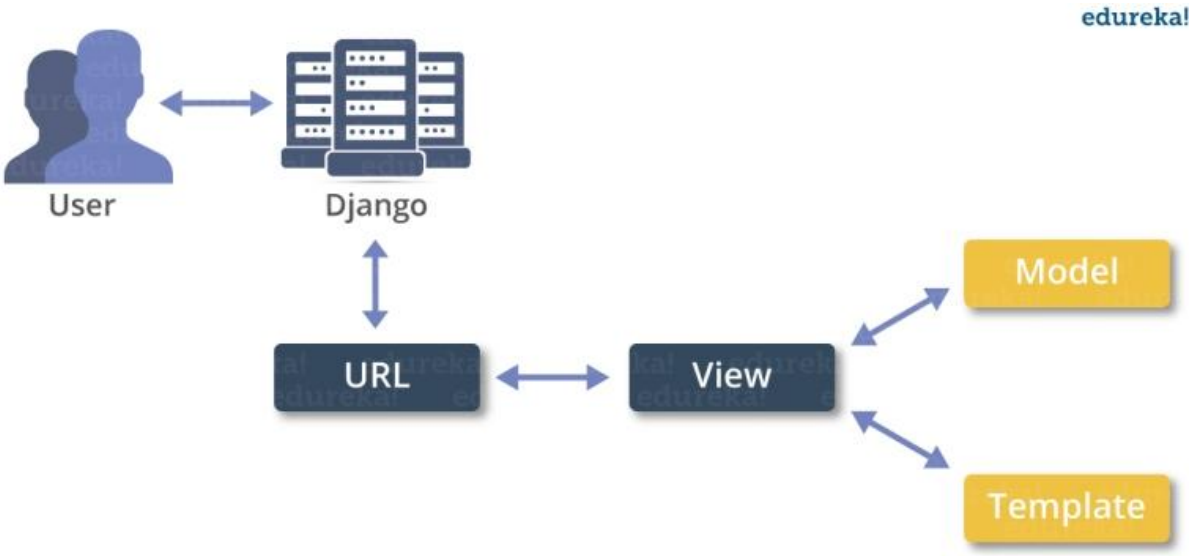
DJANGO

- Django is a high-level Python Web framework that encourages rapid development and clean, pragmatic design. Built by experienced developers, it takes care of much of the hassle of Web development, so you can focus on writing your app without needing to reinvent the wheel. It’s free and open source.
- Django's primary goal is to ease the creation of complex, database-driven websites. Django emphasizes reusabilityand "pluggability" of components, rapid development, and the principle of

don't repeat yourself. Python is used throughout, even for settings files and data models.



- Django also provides an optional administrative create, read, update and delete interface that is generated dynamically through introspection and configured via admin models



Create a Project

- Whether you are on Windows or Linux, just get a terminal or a cmd prompt and navigate to the place you want your project to be created, then use this code −
- $ django-admin startproject myproject

- This will create a "myproject" folder with the following structure −
- myproject/
- manage.py
- myproject/
- __init__.py
- settings.py
- urls.py
- wsgi.py
- The Project Structure
- The “myproject” folder is just your project container, it actually contains two elements −
- manage.py − This file is kind of your project local django-admin for interacting with your project via command line (start the development server, sync db...). To get a full list of command accessible via manage.py you can use the code −
- $ python manage.py help
- The “myproject” subfolder − This folder is the actual python package of your project. It contains four files −
- __init__.py − Just for python, treat this folder as package.
- settings.py − As the name indicates, your project settings.
- urls.py − All links of your project and the function to call. A kind of ToC of your project.
- wsgi.py − If you need to deploy your project over WSGI.
- Setting Up Your Project
- Your project is set up in the subfolder myproject/settings.py. Following are some important options you might need to set −
- DEBUG = True
- This option lets you set if your project is in debug mode or not. Debug mode lets you get more information about your project's error. Never set it to ‘True’ for a live project. However, this has to be set to ‘True’ if you want the Django light server to serve static files. Do it only in the development mode.
- DATABASES = {
- 'default': {
- 'ENGINE': 'django.db.backends.sqlite3',

- 'NAME': 'database.sql',
- 'USER': '',
- 'PASSWORD': '',
- 'HOST': '',
- 'PORT': '',
- }
- }
- Database is set in the 'Database' dictionary. The example above is for SQLite engine. As stated earlier, Django also supports −
- MySQL (django.db.backends.mysql)
- PostGreSQL (django.db.backends.postgresql_psycopg2)
- Oracle (django.db.backends.oracle) and NoSQL DB
- MongoDB (django_mongodb_engine)
- Before setting any new engine, make sure you have the correct db driver installed.
- You can also set others options like: TIME_ZONE, LANGUAGE_CODE, TEMPLATE…
- Now that your project is created and configured make sure it's working −
- $ python manage.py runserver
- You will get something like the following on running the above code −
- Validating models...
- 0 errors found
- September 03, 2015 - 11:41:50
- Django version 1.6.11, using settings 'myproject.settings'
- Starting development server at http://127.0.0.1:8000/
- Quit the server with CONTROL-C.
- A project is a sum of many applications. Every application has an objective and can be reused into another project, like the contact form on a website can be an application, and can be reused for others. See it as a module of your project.

Create an application

- We assume you are in your project folder. In our main "myproject" folder, the same folder then manage.py −

- $ python manage.py startapp myapp
- You just created myapp application and like project, Django create a “myapp” folder with the application structure −
- myapp/
- __init__.py
- admin.py
- models.py
- tests.py
- views.py
- __init__.py − Just to make sure python handles this folder as a package.
- admin.py − This file helps you make the app modifiable in the admin interface.
- models.py − This is where all the application models are stored.
- tests.py − This is where your unit tests are.
- views.py − This is where your application views are.
- Get the Project to Know About Your Application
- At this stage we have our "myapp" application, now we need to register it with our Django project "myproject". To do so, update INSTALLED_APPS tuple in the settings.py file of your project (add your app name) −
- INSTALLED_APPS = (
- 'django.contrib.admin',
- 'django.contrib.auth',
- 'django.contrib.contenttypes',
- 'django.contrib.sessions',
- 'django.contrib.messages',
- 'django.contrib.staticfiles',
- 'myapp',
-)
- Creating forms in Django, is really similar to creating a model. Here again, we just need to inherit from Django class and the class attributes will be the form fields. Let's add a forms.py file in myapp folder to contain our app forms. We will create a login form.
- myapp/forms.py

- #-*- coding: utf-8 -*-
- from django import forms
- class LoginForm(forms.Form):
- user = forms.CharField(max_length = 100)
- password = forms.CharField(widget = forms.PasswordInput())
- As seen above, the field type can take "widget" argument for html rendering; in our case, we want the password to be hidden, not displayed. Many others widget are present in Django: DateInput for dates, CheckboxInput for checkboxes, etc.
- Using Form in a View
- There are two kinds of HTTP requests, GET and POST. In Django, the request object passed as parameter to your view has an attribute called "method" where the type of the request is set, and all data passed via POST can be accessed via the request.POST dictionary.
- Let's create a login view in our myapp/views.py −
- #-*- coding: utf-8 -*-
- from myapp.forms import LoginForm
- def login(request):
- username = "not logged in"
- if request.method == "POST":
- #Get the posted form
- MyLoginForm = LoginForm(request.POST)
- if MyLoginForm.is_valid():
- username = MyLoginForm.cleaned_data['username']
- else:
- MyLoginForm = Loginform()
- return render(request, 'loggedin.html', {"username" : username})
- The view will display the result of the login form posted through the loggedin.html. To test it, we will first need the login form template. Let's call it login.html.
- <html>
- <body>
- <form name = "form" action = "{% url "myapp.views.login" %}"
- method = "POST" >{% csrf_token %}

```
        <div style = "max-width:470px;">
          <center>
            <input type = "text" style = "margin-left:20%;"
              placeholder = "Identifiant" name = "username" />
          </center>
        </div>
        <br>
        <div style = "max-width:470px;">
          <center>
            <input type = "password" style = "margin-left:20%;"
              placeholder = "password" name = "password" />
          </center>
        </div>
        <br>
        <div style = "max-width:470px;">
          <center>
            <button style = "border:0px; background-color:#4285F4; margin-top:8%;
              height:35px; width:80%;margin-left:19%;" type = "submit"
              value = "Login" >
              <strong>Login</strong>
            </button>
          </center>
        </div>
      </form>
    </body>
</html>
```

- The template will display a login form and post the result to our login view above. You have probably noticed the tag in the template, which is just to prevent Cross-site Request Forgery (CSRF) attack on your site.
- {% csrf_token %}

- Once we have the login template, we need the loggedin.html template that will be rendered after form treatment.
- <html>
- <body>
- You are : {{username}}
- </body>
- </html>
- Now, we just need our pair of URLs to get started: myapp/urls.py
- from django.conf.urls import patterns, url
- from django.views.generic import TemplateView
- urlpatterns = patterns('myapp.views',
- url(r'^connection/',TemplateView.as_view(template_name = 'login.html')),
- url(r'^login/', 'login', name = 'login'))
- When accessing "/myapp/connection", we will get the following login.html template rendered −
- Setting Up Sessions
- In Django, enabling session is done in your project settings.py, by adding some lines to the MIDDLEWARE_CLASSES and the INSTALLED_APPS options. This should be done while creating the project, but it's always good to know, so MIDDLEWARE_CLASSES should have −
- 'django.contrib.sessions.middleware.SessionMiddleware'
- And INSTALLED_APPS should have −
- 'django.contrib.sessions'
- By default, Django saves session information in database (django_session table or collection), but you can configure the engine to store information using other ways like: in file or in cache.
- When session is enabled, every request (first argument of any view in Django) has a session (dict) attribute.
- Let's create a simple sample to see how to create and save sessions. We have built a simple login system before (see Django form processing chapter and Django Cookies Handling chapter). Let us save the username in a cookie so, if not signed out, when accessing our login page you won’t see the login form. Basically, let's make our login system we used in Django Cookies handling more secure, by saving cookies server side.
- For this, first lets change our login view to save our username cookie server side −

- def login(request):
- username = 'not logged in'
- if request.method == 'POST':
- MyLoginForm = LoginForm(request.POST)
- if MyLoginForm.is_valid():
- username = MyLoginForm.cleaned_data['username']
- request.session['username'] = username
- else:
- MyLoginForm = LoginForm()
-
- return render(request, 'loggedin.html', {"username" : username}
- Then let us create formView view for the login form, where we won't display the form if cookie is set −
-
- def formView(request):
- if request.session.has_key('username'):
- username = request.session['username']
- return render(request, 'loggedin.html', {"username" : username})
- else:
- return render(request, 'login.html', {})
- Now let us change the url.py file to change the url so it pairs with our new view −
-
- from django.conf. urls import patterns, url
- from django.views.generic import TemplateView
- urlpatterns = patterns('myapp.views',
- url(r'^connection/','formView', name = 'loginform'),
- url(r'^login/', 'login', name = 'login'))
- When accessing /myapp/connection, you will get to see the following page

CHAPTER-6 SYSTEM STUDY

6.1 FEASIBILITY STUDY

The feasibility of the project is analyzed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

Three key considerations involved in the feasibility analysis are,

- **ECONOMICAL FEASIBILITY**
- **TECHNICAL FEASIBILITY**
- **SOCIAL FEASIBILITY**

6.1.1 ECONOMICAL FEASIBILITY

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

6.1.2 TECHNICAL FEASIBILITY

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands on the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

6.1.3 SOCIAL FEASIBILITY

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system.

CHAPTER-7 SYSTEM TESTING

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub assemblies, assemblies and/or a finished product It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

7.1 TYPES OF TESTS

7.1.1 Unit testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application .it is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

7.1.2 Integration testing

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfaction, as shown by successfully unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

7.1.3 Functional testing

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centered on the following items:

Valid Input : identified classes of valid input must be accepted.

Invalid Input : identified classes of invalid input must be rejected.

Functions : identified functions must be exercised.

Output : identified classes of application outputs must be exercised.

Systems/Procedures : interfacing systems or procedures must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

7.1.4 System Testing

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

White Box Testing

White Box Testing is a testing in which in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is purpose. It is used to test areas that cannot be reached from a black box level.

Black Box Testing

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other

kinds of tests, must be written from a definitive source document, such as specification or requirements document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box .you cannot "see" into it. The test provides inputs and responds to outputs without considering how the software works.

7.1.5 Integration Testing

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects.

The task of the integration test is to check that components or software applications, e.g. components in a software system or – one step up – software applications at the company level – interact without error.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

7.1.6 Acceptance Testing

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

Test strategy and approach

Field testing will be performed manually and functional tests will be written in detail.

Test objectives

- All field entries must work properly.
- Pages must be activated from the identified link.
- The entry screen, messages and responses must not be delayed

Features to be tested

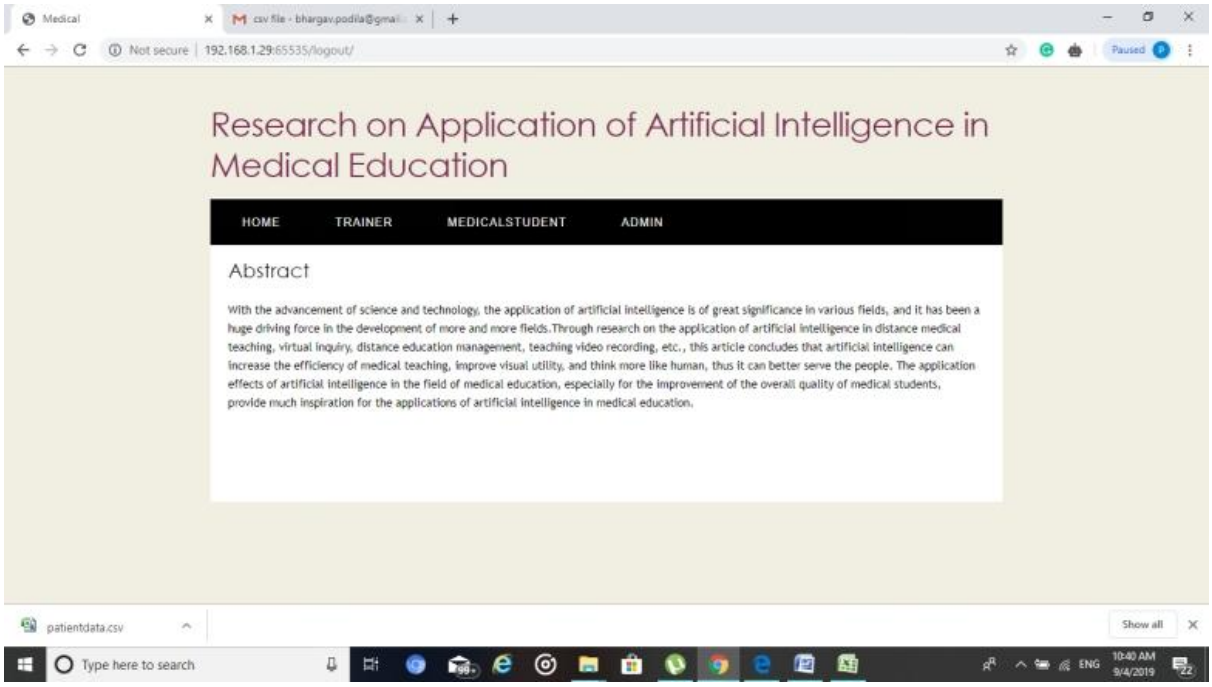
- Verify that the entries are of the correct format
- No duplicate entries should be allowed
- All links should take the user to the correct page.

7.2 TEST CASES

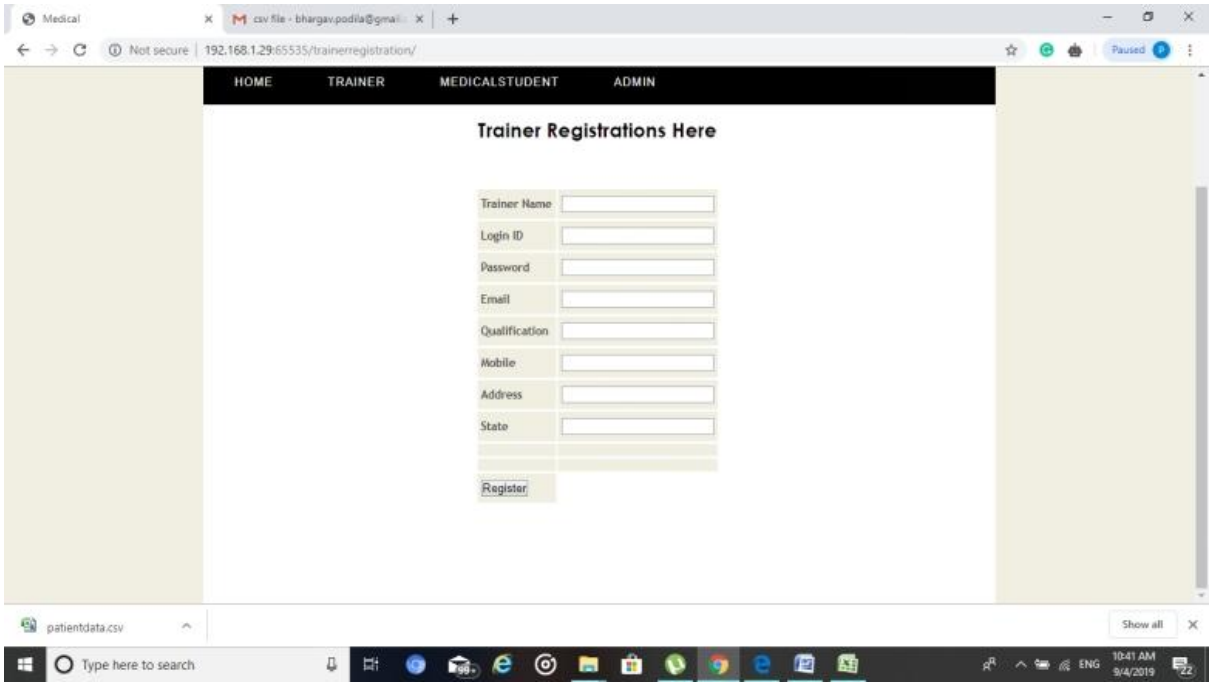
S.no	Test Case	Excepted Result	Result	Remarks(IF Fails)
1.	TRAINER REGISTERED	If TRAINER registration successfully.	Pass	If TRAINER is not registered.
2.	TRAINER LOGIN	If TRAINER name and password is correct then it will getting valid page.	Pass	If TRAINER name or password is not correct.
3.	MEDICAL STUDENT REGISTRATION	If MEDICAL STUDENT is registered successfully.	Pass	If MEDICAL STUDENT is not registered.
4.	MEDICAL STUDENT LOGIN	MEDICAL STUDENT name and password is correct then getting valid page.	Pass	If MEDICAL STUDENT name or password is wrong.
5.	ADMIN	TRAINER rights and MEDICAL STUDENT rights will be accepted here.	Pass	If TRAINER and MEDICAL STUDENT are not registered.
6.	TRAINER UPLOAD DATA	Choose or select MEDICAL DATA and upload in ADMIN.	Pass	If TRAINER is not select or upload the DATA.
7.	VIEW MEDICAL STUDENT DATAS	DOCTOR view MEDICAL STUDENT DATA details are available here	Pass	If MEDICAL STUDENT DATA is not uploaded in ADMIN.
8.	MEDICAL STUDENT VIEW DATAS AND VIDEOS	TRAINER uploaded DATA AND VIDEOS details are available here and we can view VIDEOS	Pass	If DATA AND VIDEOS is not uploaded in ADMIN.

CHAPTER-8 RESULTS

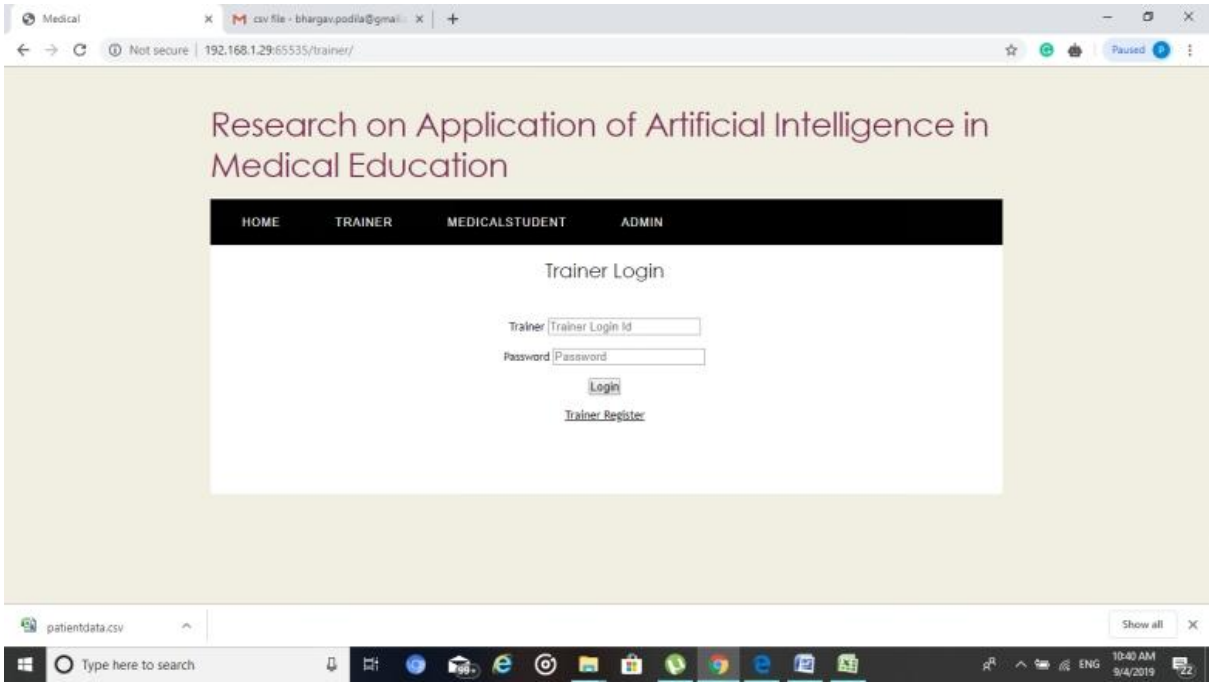
HOME PAGE:



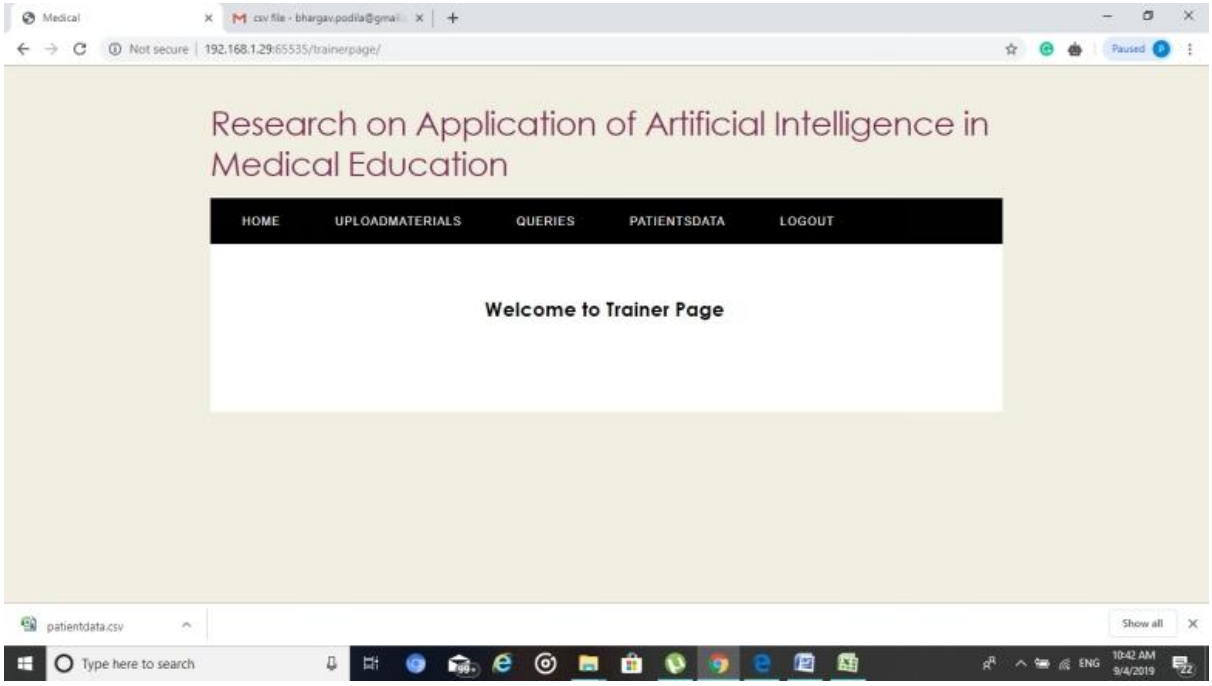
TRAINER REGISTER:



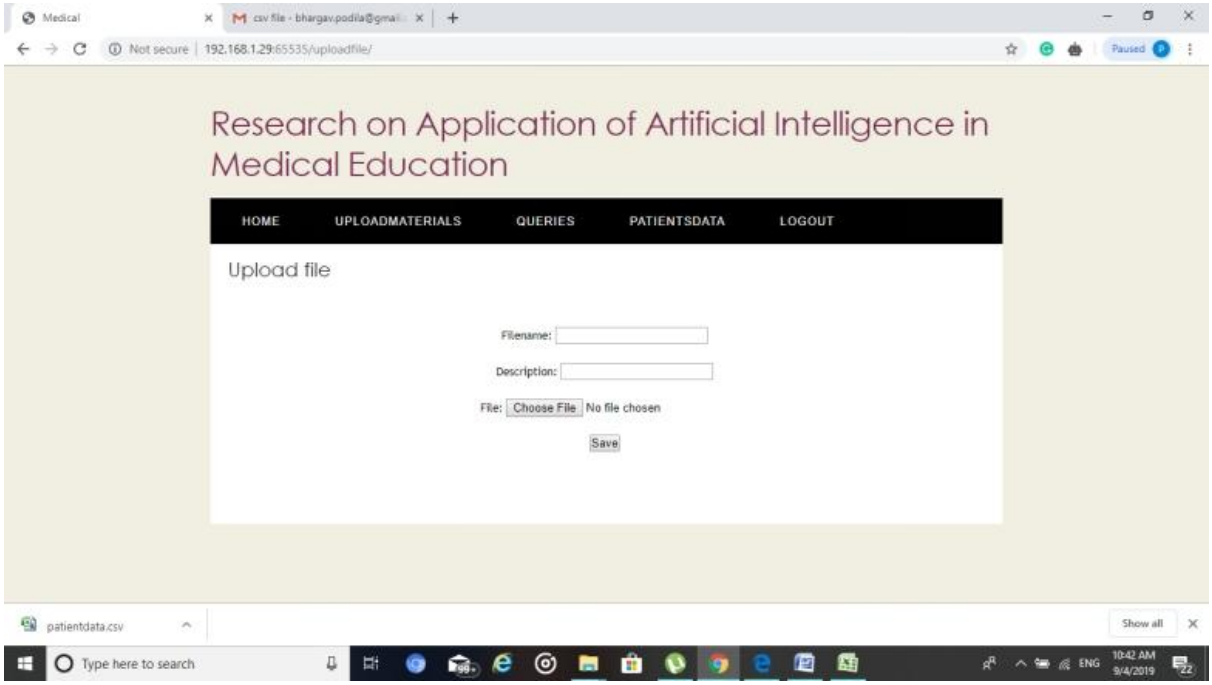
TRAINER LOGIN:



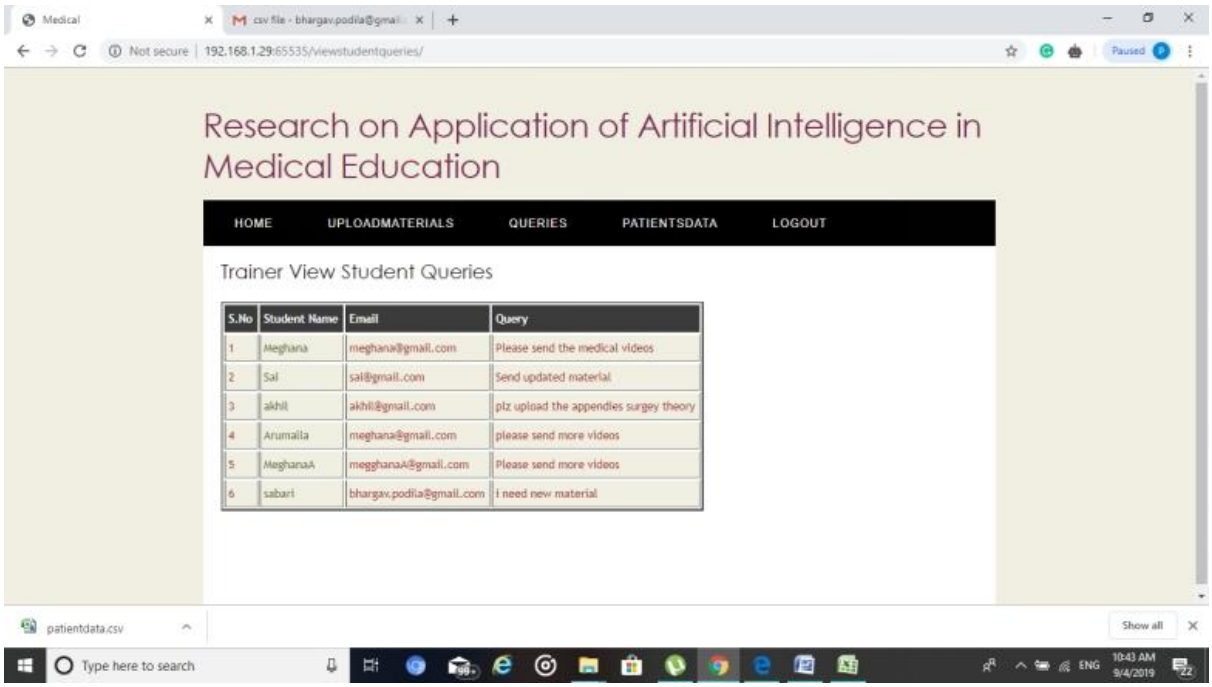
TRAINER HOME PAGE:



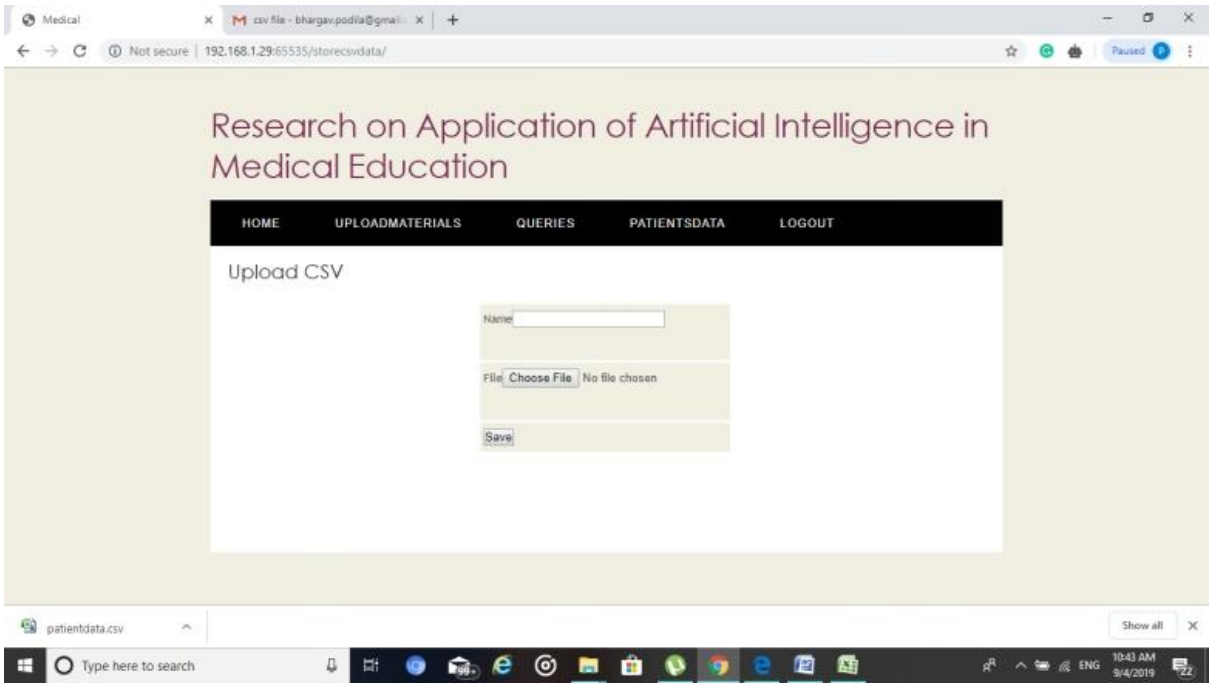
UPLOAD TRAINER FILES:



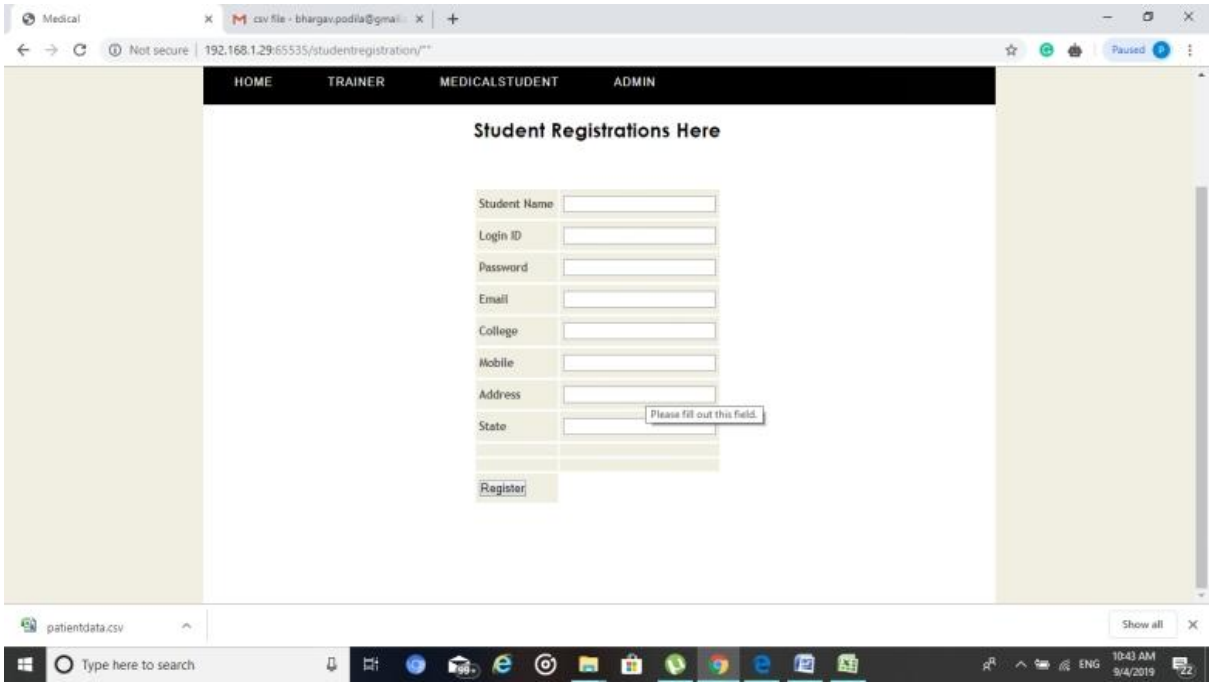
TRAINER VIEW STUDENT QUERIES:

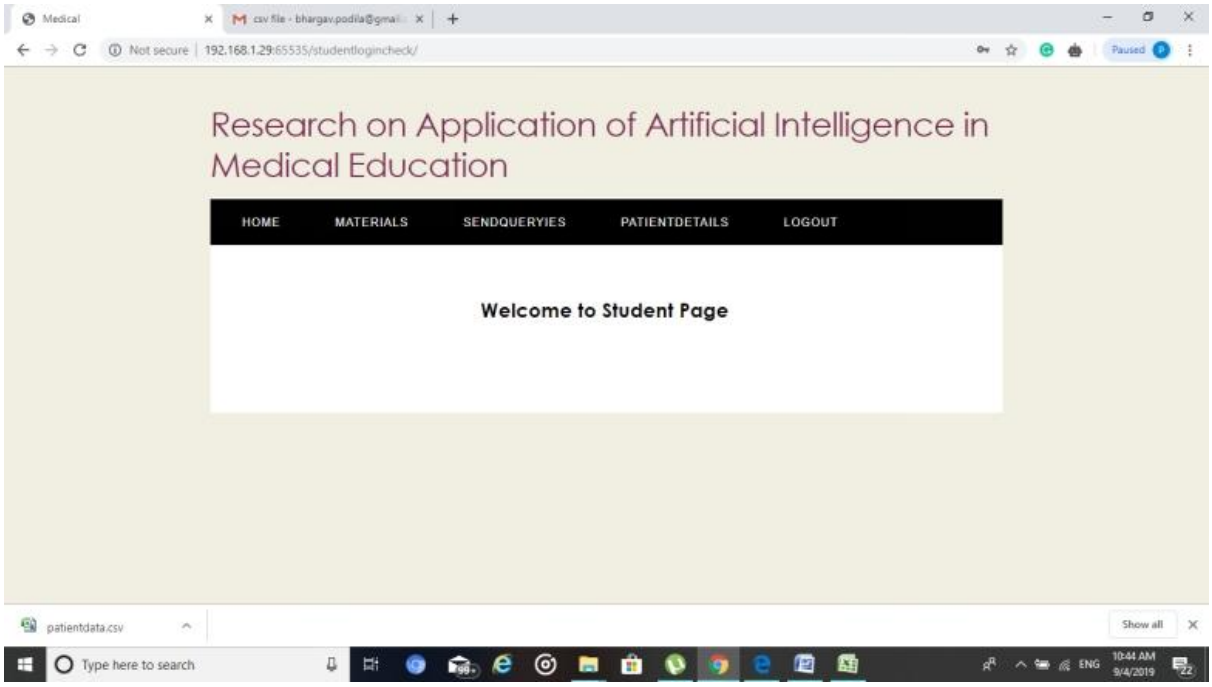


UPLOAD CSV:

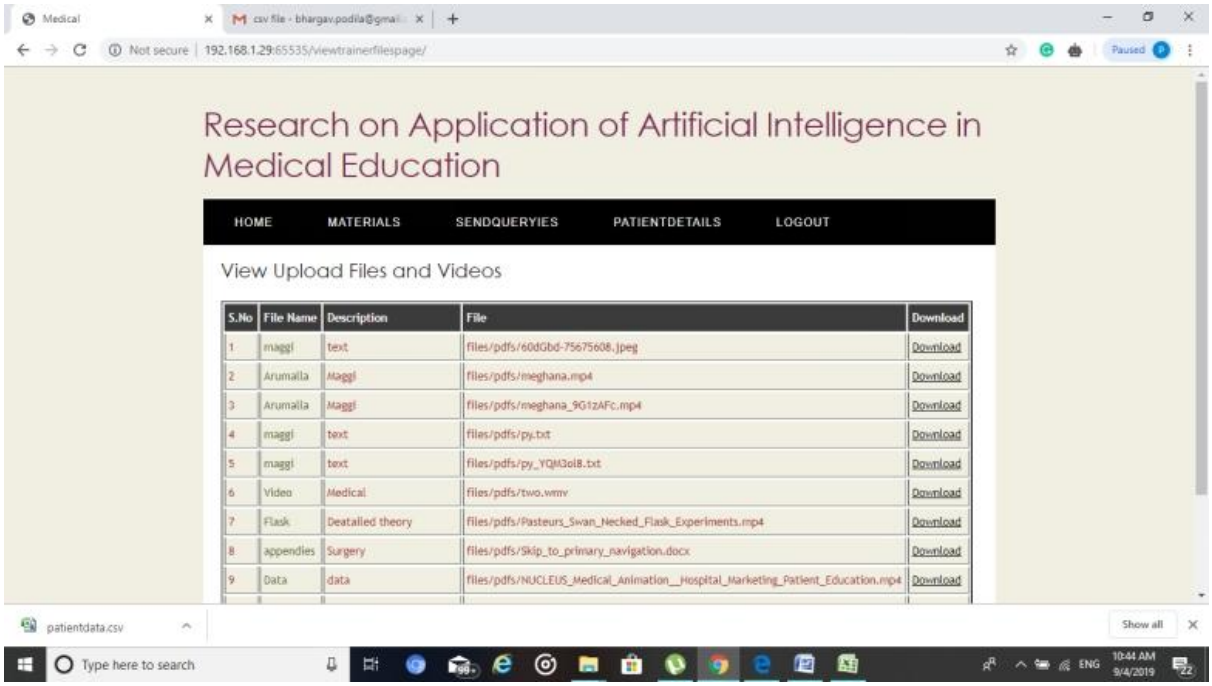


STUDENT REGISTRACTIONS:

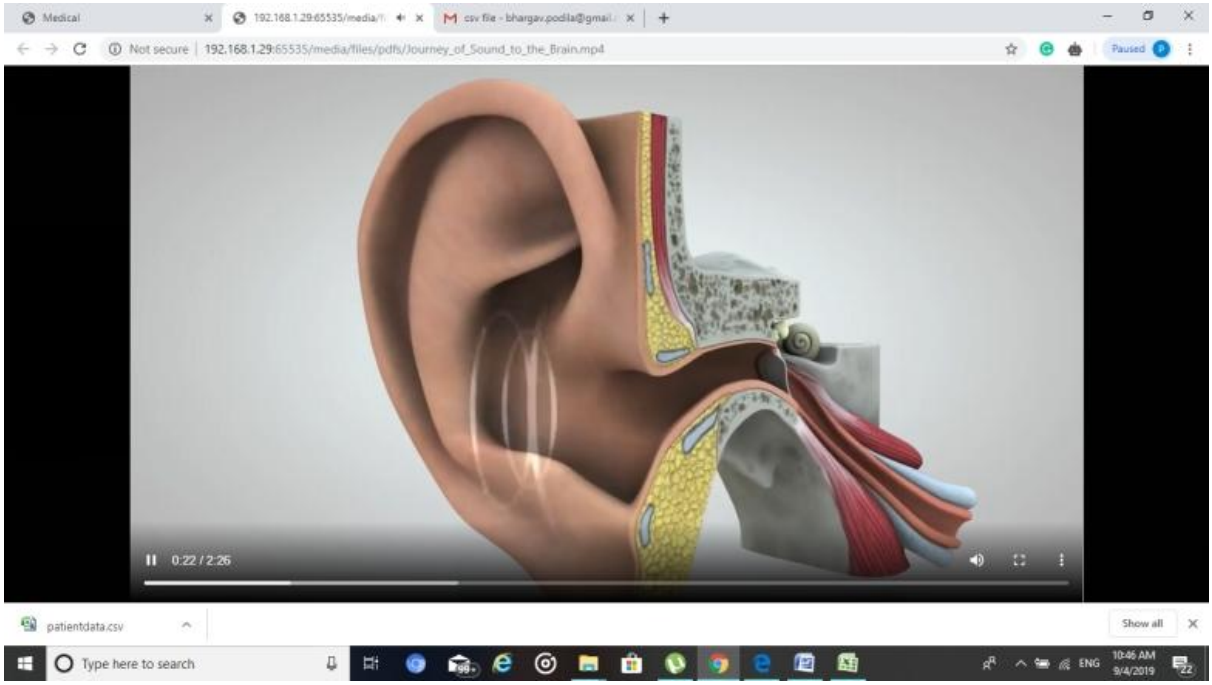




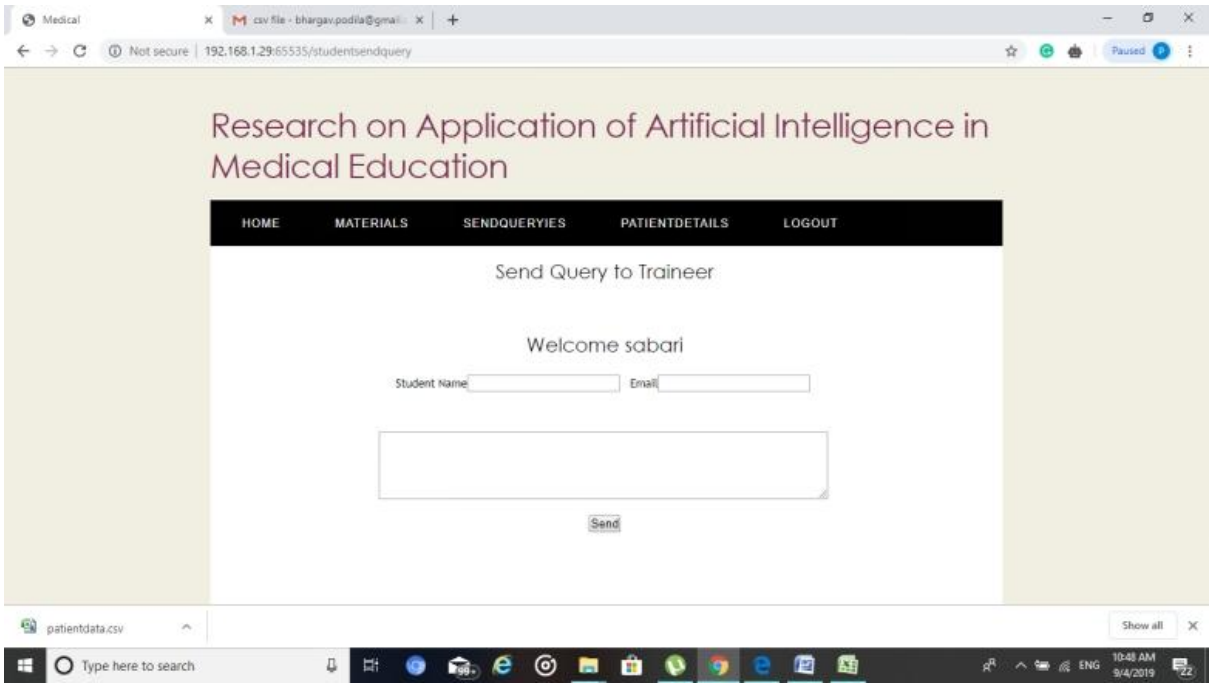
VIEW UPLOAD FILES:



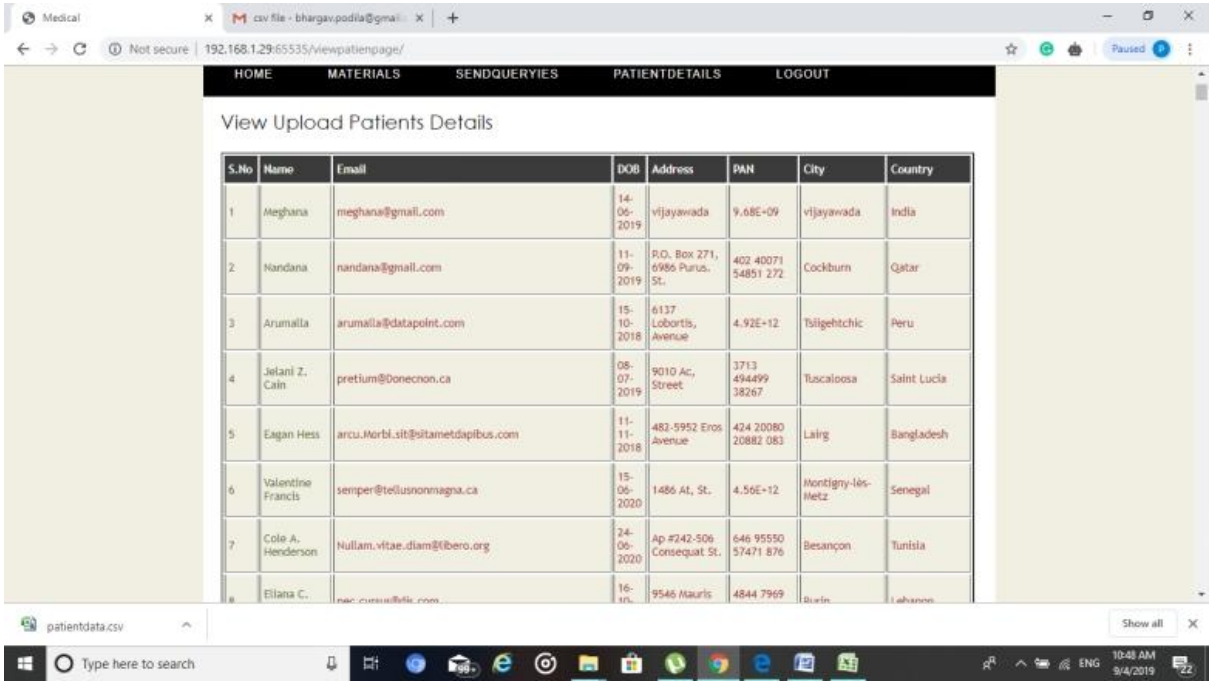
DOWNLOAD VIDEO:



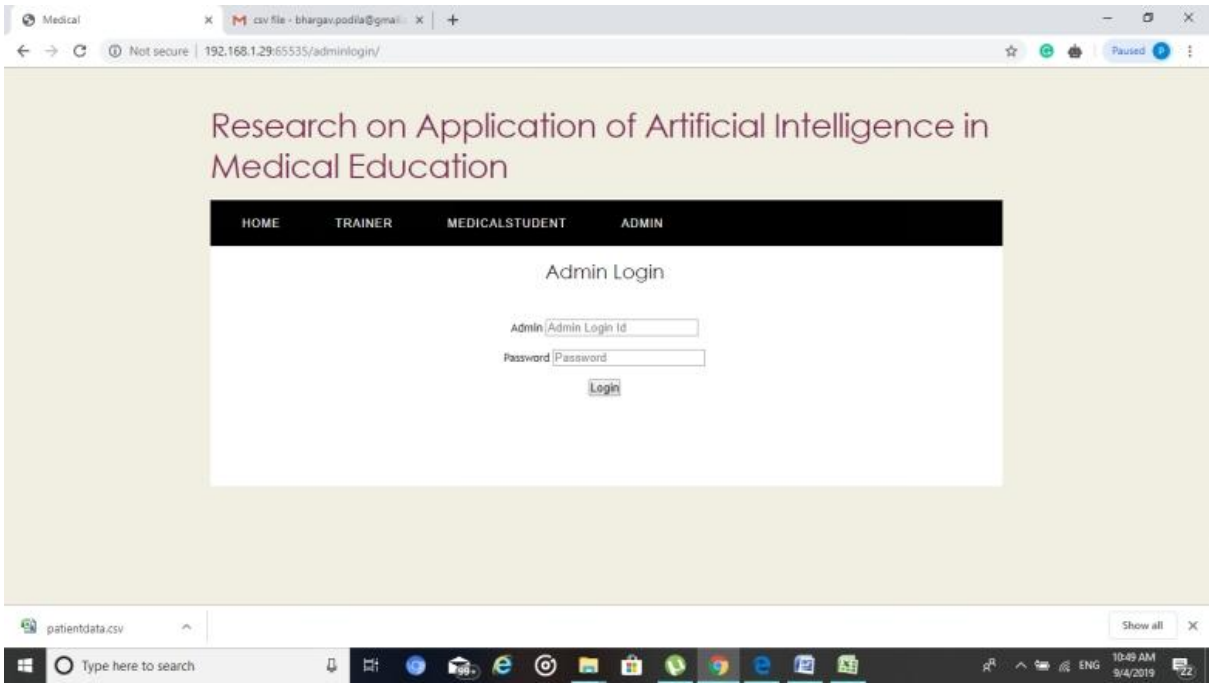
SEND QUERY TO TRAINEER:

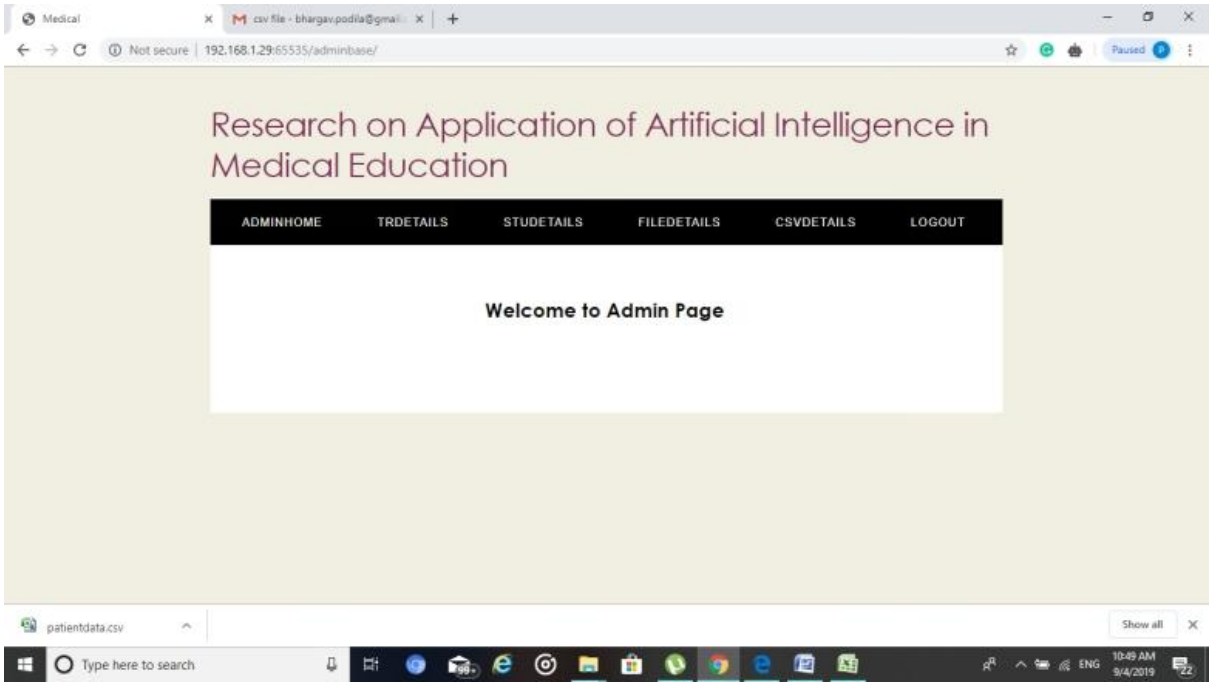


VIEW UPLOAD PATIENT DETAILS:

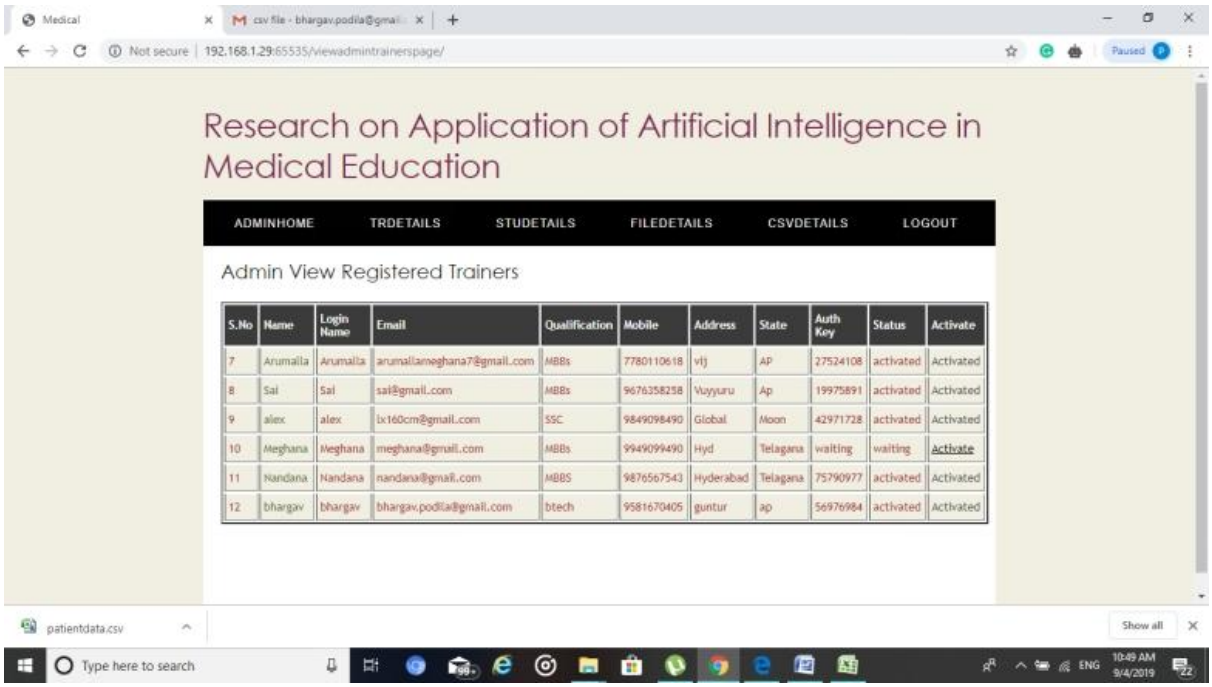


ADMIN LOGIN:

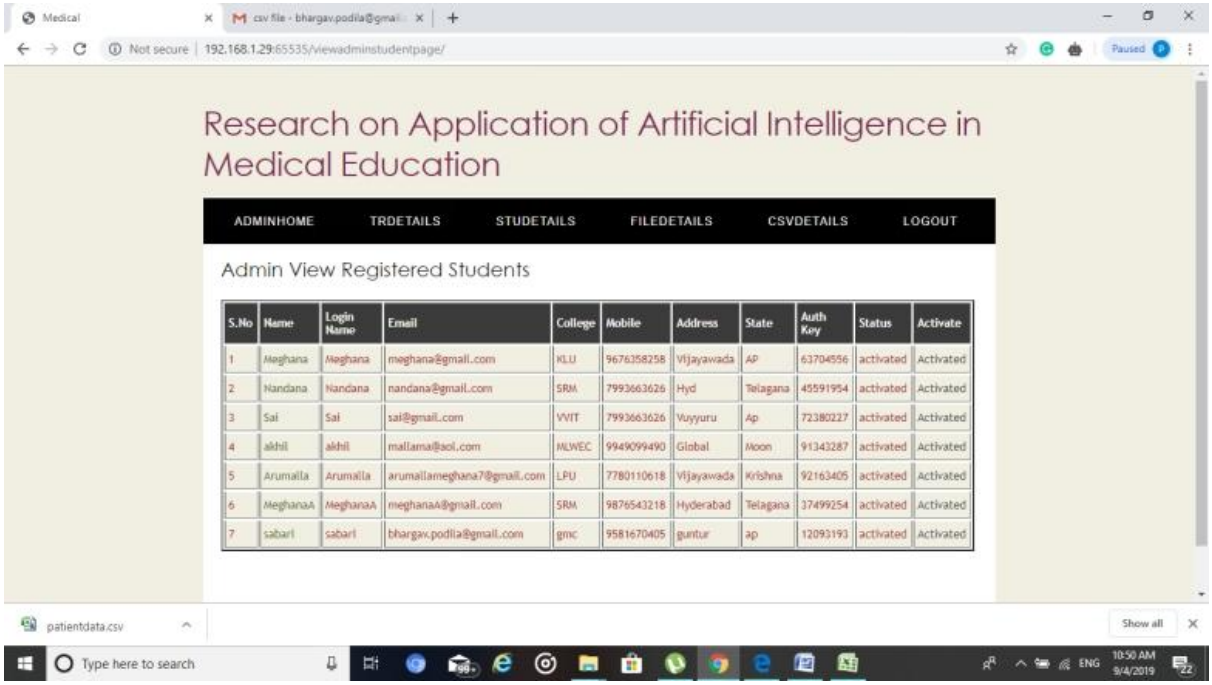




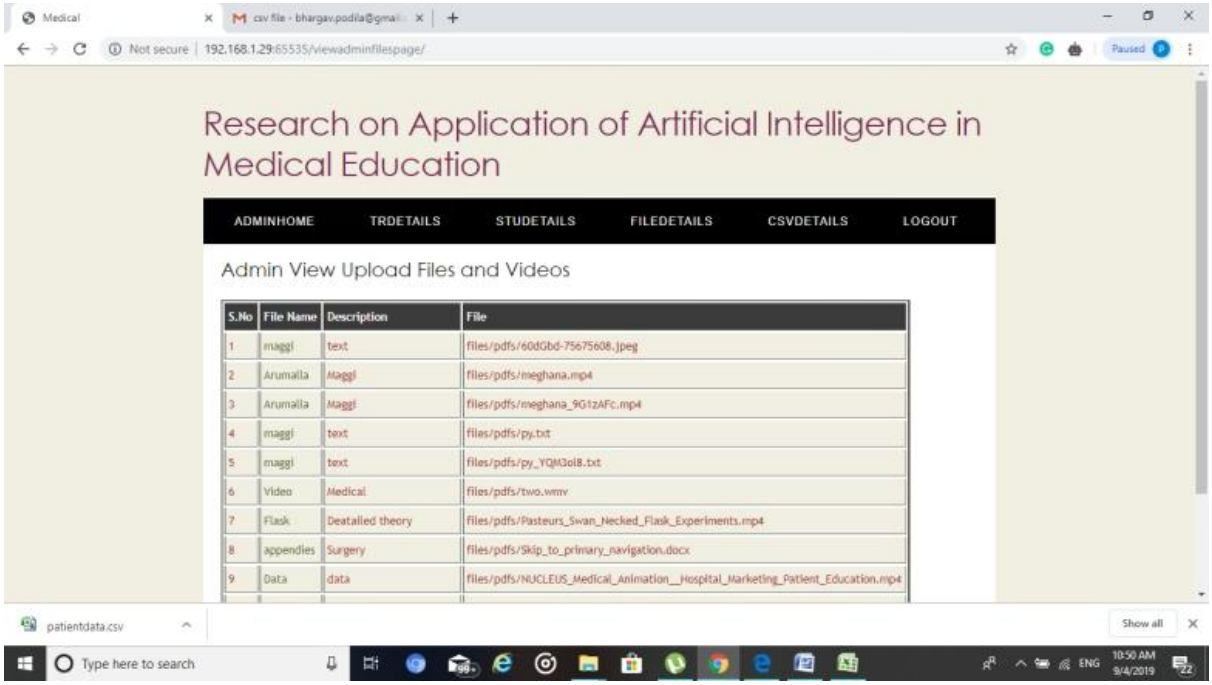
ADMIN VIEW REGISTERED TRAINERS:



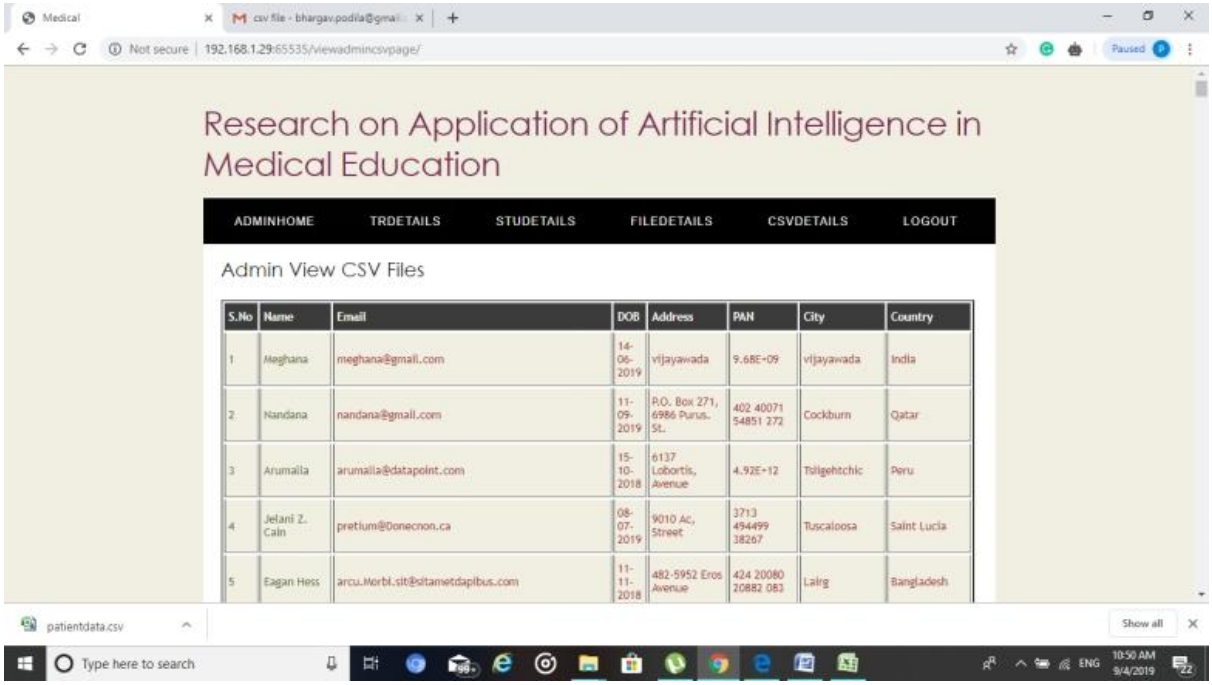
ADMIN VIEW REGISTERED STUDENTS:



VIEW UPLODED FILES:



VIEW PATIENT CSV FILE:



CHAPTER 9 CONCLUSION

The essence of education is accumulation and inheritance, inheriting the knowledge accumulated by the predecessors to future generations and encouraging them to innovate through educational means. The fundamental of artificial intelligence technology is to accumulate knowledge through machine learning, artificial neural network, data mining and other methods. Through decision supports, the expert system spreads knowledge and applies it. This article analyzes the changes in the way that artificial intelligence technology modifies traditional medical education. A key way that artificial intelligence affects medical education is to support personalized learning, help students at different levels, and provide help and support when teachers and students need it. Artificial intelligence can not only help teachers and students design courses that meet their needs, but also can focus on student performance and alert teachers when problems may arise, helping teachers improve teaching methods. Artificial intelligence will change the role of teachers. Teachers will supplement artificial intelligence courses to provide students with interpersonal interaction and practical experience. Using artificial intelligence systems, students can learn anytime, anywhere, and some classroom teaching can be substituted by these programs.

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CHAPTER -11 YUKTHI INNOVATION CERTIFICATE